

Where to Measure, What to Measure With

Evidence-Aware Drinking-Water Monitoring Under Uncertainty

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Introduction — why this matters

363 million people in India lack safely managed drinking water. [1]
35 / 38 surveys found fewer households free of *E. coli* at point of use than at collection; mean **-12.9 pp**. [2]

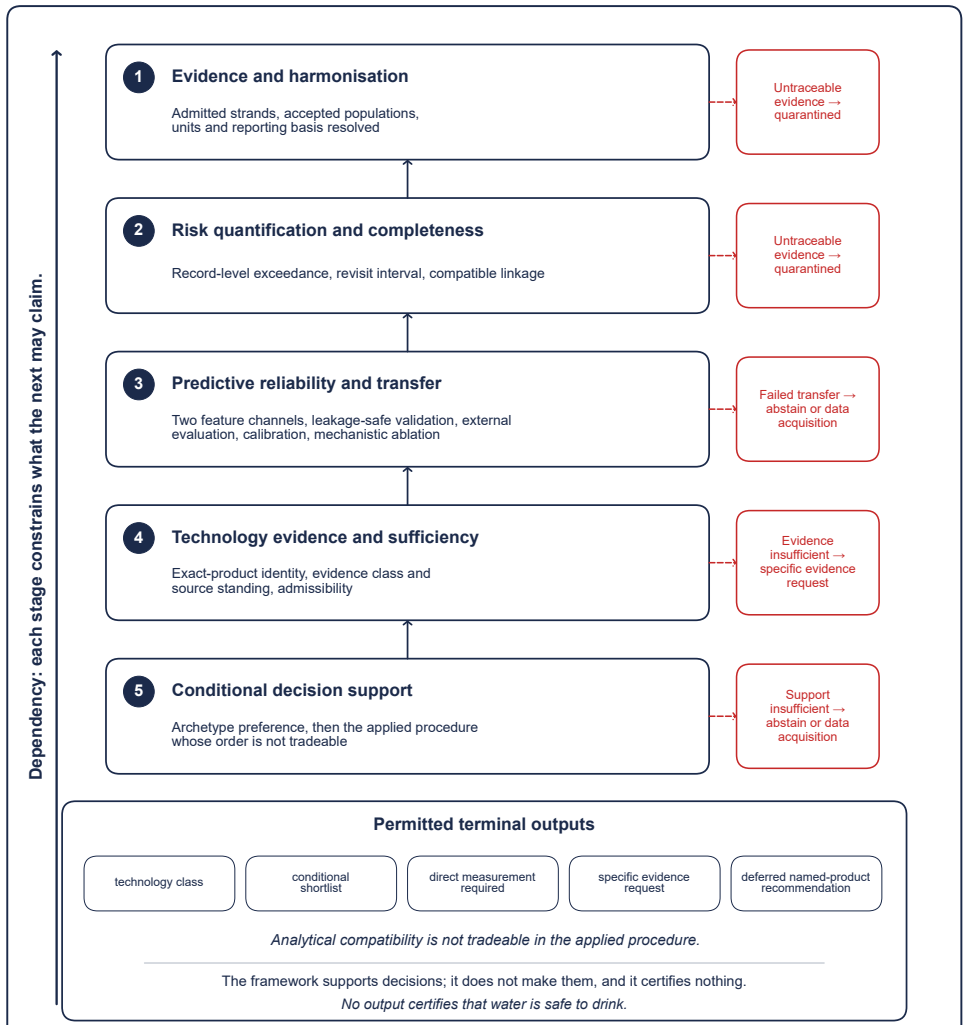
Where to test, and what to test with?

Literature review — what was missing

DATA units, chemical basis and denominators differ across archives. **PREDICTION** random validation over-states new-area performance. **TECHNOLOGY** nominal capability is not exact-product evidence.

Gap: no study connected all three into a pathway permitted to stop and request evidence. [3–5]

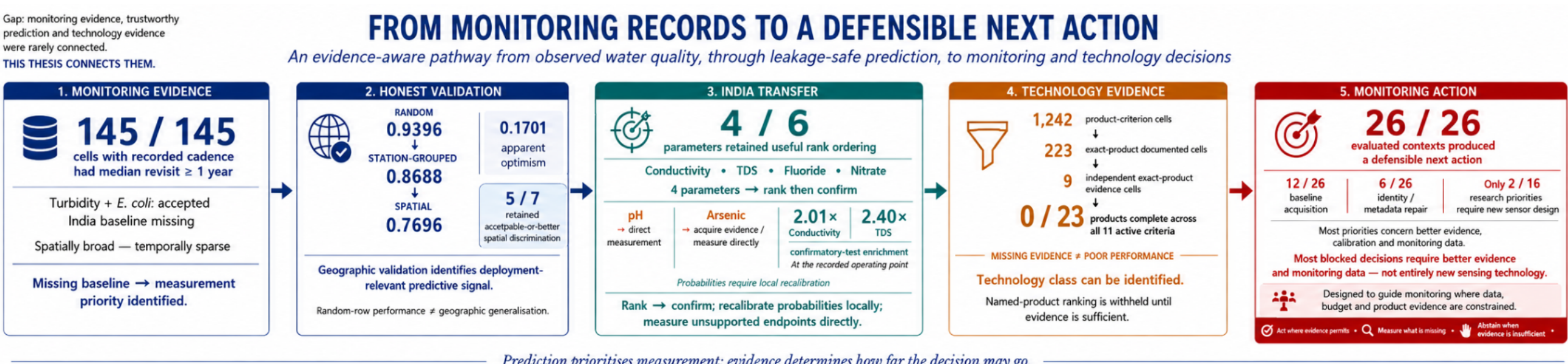
Methodology



Five-stage evidence-to-decision sequence; abstention is a designed output. Source: thesis Fig. 3.1.

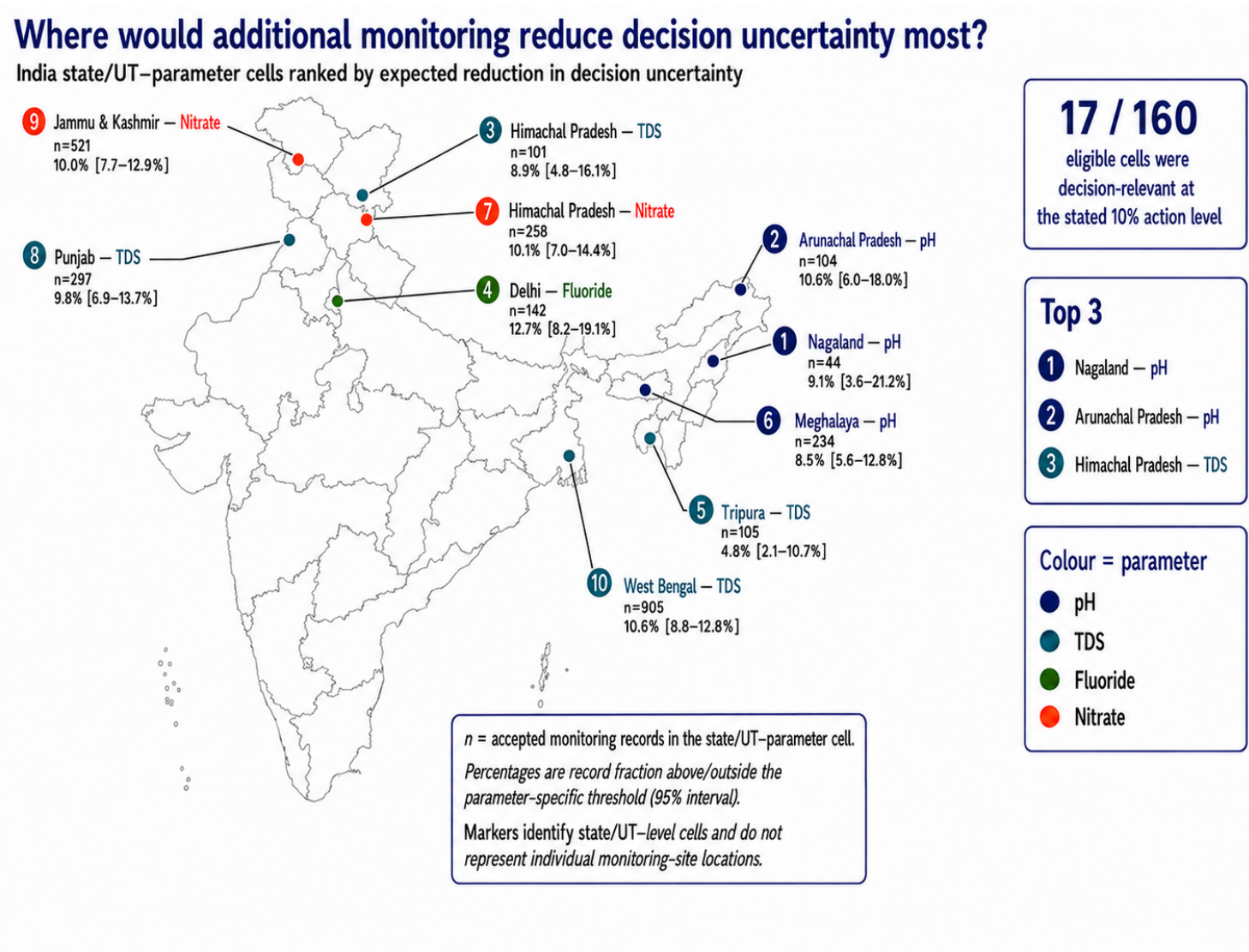
References

[1] WHO & UNICEF (2023). *Progress on household drinking water, 2000–2022*.
[2] Santos, T. *et al.* (2023). *E. coli* in drinking water sources. *J. Water Health*.
[3] Roberts, D. *et al.* (2017). Cross-validation for structured data. *Ecography*.
[4] Camargo, E. *et al.* (2023). Low-cost water sensors for IoT. *Sensors*.
[5] Pichel Mira, N. *et al.* (2023). Field testing in LMIC regions. *J. Environ. Chem. Eng.*
[6] Trimmer, J. *et al.* (2024). Costs of monitoring in rural Africa. *Environ. Sci. Technol.*
[7] Van Geen, A. *et al.* (2019). Testing 30,000 wells for arsenic. *Sci. Total Environ.*



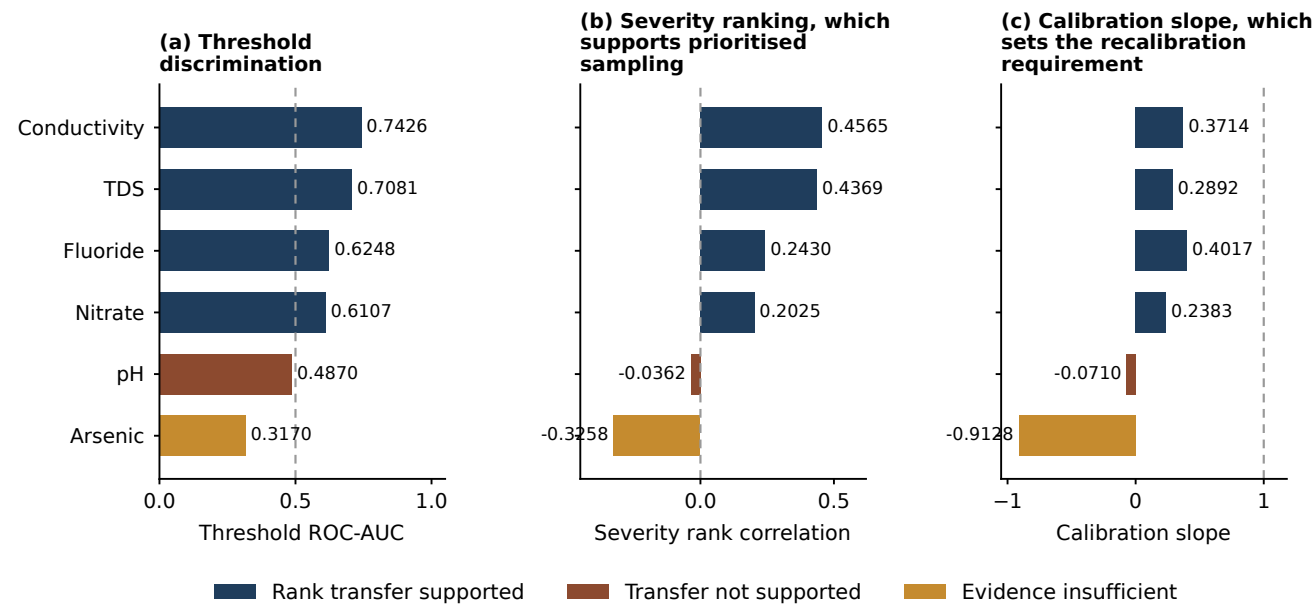
Thesis-derived synthesis; values verified against the submitted report.

Results & Discussion — what survived, what transferred, and where measurement matters next



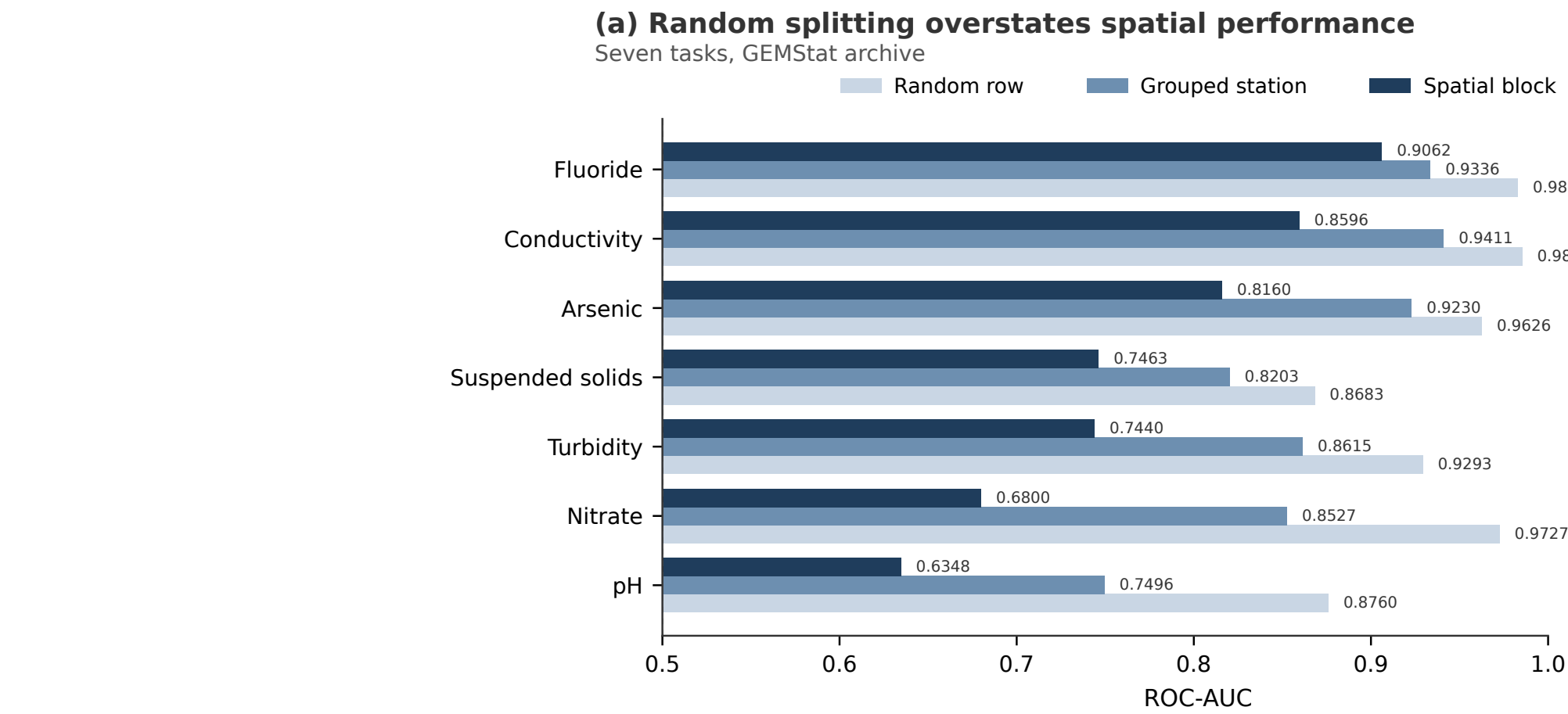
State/UT-level measurement priority, not contamination severity or population prevalence. Source: Fig. 4.7 / Table 4.11.

External India transfer: ranking survives where calibration does not



Conductivity, TDS, fluoride and nitrate retain useful ordering; pH routes to direct measurement, arsenic to evidence acquisition. Local recalibration precedes probability-based use. Source: Fig. 4.5 / Table 4.6.

Honest validation: what survives geographic separation?



5 / 7 tasks retain acceptable-or-better spatial discrimination; label shuffle collapses to **ROC-AUC 0.501**. **Random splitting overstated deployment-facing performance; useful signal remained for most tasks.** Source: submitted thesis, Fig. 4.4 panel (a), Tables 4.3–4.4 and 4.7.

From analysis to monitoring action

Rajasthan: conductivity 12,349 records, **54.06%** above the 1,500 $\mu\text{S}/\text{cm}$ convention; TDS 3,131 records, **51.96%** above 1,000 mg/L .

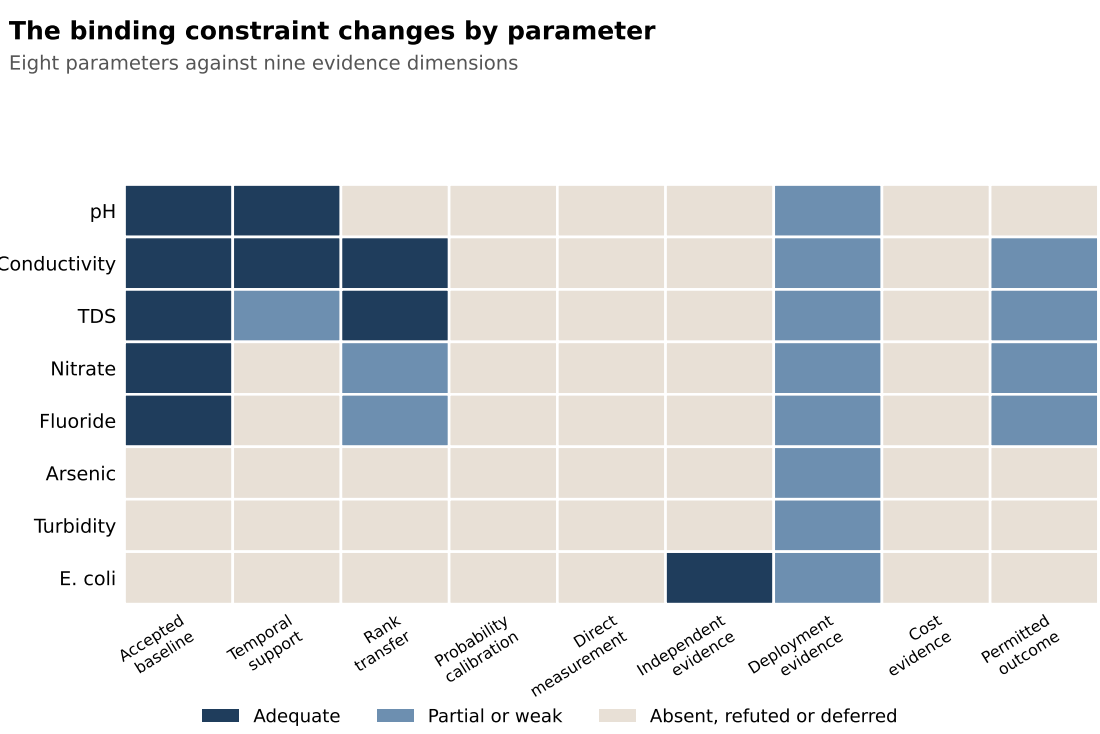
Rank → confirm → recalibrate locally → technology-class shortlist → product-level abstention

Only 2 / 16 research priorities concern sensor design.

Most need evidence existing classes can supply: threshold-near validation • recalibration labels • cost and energy per valid result • reliability • operator effect • persistent identifiers. [6, 7]

1. Honest validation preserves decision-useful signal.
2. Rank transfer supports targeted confirmation, 4 of 6.
3. All 26 contexts return a defined next action.

Prediction prioritises measurement; evidence sets how far the decision may go.



Data constrain arsenic, turbidity and *E. coli*; pH at transfer; the four rank-supported parameters at calibration and product evidence. Source: Fig. 4.13.